

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

SUMMER SEMESTER, 2020-2021

DURATION: 3 HOURS

FULL MARKS: 150

CSE 4849: Human Computer Interaction

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

Based on the following scenario and story, answer question number 1, 2, 3 and 4.

Restaurant 'Seasonal Platter' is a special kind of restaurant. During different seasons, like summer, winter or fall, they offer different types of specialty dishes. A special software is used to offer the dishes to the *customers for their choice*. At the end of the season, the customers who have an account with the restaurant can upvote or downvote the dishes, which would then affect the dishes of the later seasons. Dishes are actually *offered by the chefs themselves*, since they are responsible for offering their own specialty. A *manager oversees the whole process* and can modify, update or remove a dish offered by any chef.

Upon interviewing a famous chef, Mr. Rordon Gamsey from 'Seasonal Platter', a description of the overall process of offering a new dish was found and is attached below.

"Yeah, I remember the whole process. It is a bit tedious, but nothing too complex. First you need to log in with your chef account. Once there, you can offer new dishes. But before you offer, first you got to check whether your dish already exists or not. Like my Pineapple Smoothie is an all season favorite and it is there, always. So I do not need to re-offer it every time. However they did not like my Chicken Pot Pie last season, so it got removed from the display automatically by the system. I am offering it this season again though, since I believe it deserves a second chance. Since it was already offered once before, I just need to put it in the display using a single click, nothing an old chef cannot handle. I am also offering Chocolate Naga Cake this season, a completely new recipe. For this one though, I first need to create a new dish, put the name and the list of all the ingredients etc. Like you know, people have personal preferences, right? Once the dish is created, I can offer it to the whole world by a single click."

The owner of 'Season Platter' has decided to integrate an interactive display in each of the tables of the restaurant so that the customers can easily browse the dishes, order and register themselves and give their feedback. He has also decided to put the systems in the kitchen so that the chefs can fully utilize their eureka moments while inventing new dishes and register them right away.

1. a) With respect to the scenario, extract 3 requirements for each of the following cases that could have been used to build the system. Give a 1 or 2 line justification for each of them.
 - i. Functional Requirements
 - ii. Data Requirements
 - iii. Environmental Requirements
 - iv. User Requirements
 - v. Usability Requirements

5□3=15
 (CO4)
 (PO2)

- b) Based on the user story written above, create a Hierarchical Task Analysis with Plans and Graphs for *offering a new/existing dish by a chef*. 10
(CO4)
(PO2)
2. a) Briefly elaborate on two data gathering methods that you would find suitable in the above context if you were asked to implement the system from scratch. For each of the methods, explain how the specific method's advantage is going to fit the system and how the disadvantage needs to be overcome. $2 \times 4 = 8$
(CO4)
(PO2)
- b) What are the different personas that can be taken into consideration while designing the aforementioned system based on the three roles, customers, chef and manager? Mention 6 personas with just their abilities or shortcomings with respect to the system in 2-3 sentences. You can choose your own distribution of persona under each role, for instance, 3 personas under customers, 2 under chefs and 1 under manager etc. $6 \times 2 = 12$
(CO4)
(PO2)
- c) Can the process of affinitization help the development of the aforementioned system? Give proper reasoning behind your answer within 10 simple sentences. 5
(CO4)
(PO2)
3. a) Different prototyping techniques are suitable for different tasks. For instance, the above story can be represented using storyboarding or a series of sketches. Now, for the following tasks, mention a suitable prototyping method with 3 sentences explaining each why they are suitable. $3 \times 3 = 9$
(CO5)
(PO2)
- Creating UI for the chefs for offering recipe
 - Describing the interaction sequence of voting to the users
 - Taking feedback from the users on the system
- b) Represent a chef offering a new dish using storyboarding. The drawings need not be very good, but clear and understandable drawings are required. 10
(CO5)
(PO2)
- c) Explain the sentence, "Though prototyping might prove to be costly in the initial phase, it may save valuable time and resources in the long run". 6
(CO1)
(PO1)
4. a) With respect to the scenario and aforementioned questions, illustrate how the Star Model can facilitate a user centric design process. 5
(CO3)
(PO2)
- b) Suppose the owner of the restaurant wants to find an answer to the research question, "Does the color of the customer's UI influence the purchase decision and amount of a specific type of food (appetizer, main course and/or dessert)?" With respect to the research question, $5 + 10 + 5 = 20$
(CO5)
(PO2)
- Describe an experiment with steps that may facilitate the answer to the question.
 - Formally construct the independent, dependent, control, random and confounding variables of the experiment.
 - Choose and justify a proper evaluation approach for the experiment.
5. Suppose you want to design a command line application for a certain task. The application has upto 30 configurable parameters. In your current design, the user may pass the value of each of the arguments or parameters while calling it from the terminal. Though 'help' can be called to list down all the possible parameters, novice users still find it difficult to properly use the application. And in many cases, for a certain user or environment, a lot of the parameters remain the same. For the simplicity of understanding, invocation of a command line tool, *gcc* for compiling a single C file is given below.

```
$ gcc Filename.c -o Application
```

The program takes 3 arguments, first one being the file name to compile, then '-o' to indicate that the next argument would be output file name/path and then accordingly, the output file name. Now based on this scenario, answer the following questions.

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|-------|--|----------------------|
| a) | Point out how human memory, both short term and long term, can affect the usability of the aforementioned application. | 8
(CO2)
(PO2) |
| b) | Propose an ideal solution (a single one based on your decision) to the problem of having 30 parameters in the aforementioned program. You cannot decrease the number of parameters | 9
(CO2)
(PO2) |
| c) | Exemplify how the characteristics of reading text of human beings should influence the design of terminals and the output of command line programs. | 8
(CO2)
(PO2) |
| 6. a) | Based on your understanding from the classes, define what HCI is and where it is used under 15 simple sentences. | 10
(CO1)
(PO1) |
| b) | Note down and describe at least 3 domains that you think contributed in making this question paper. It was written in Google Docs. Each domain description should not exceed more than 5 simple sentences. | 8
(CO1)
(PO1) |
| c) | Define interface metaphors and how they influence the design of user interface in computer science under 7 sentences. | 7
(CO1)
(PO1) |